

Chapter 2

Gravitational Acceleration

The Earth's gravitational acceleration is derived from the universal law of gravitation. In principle, any object with mass will always attract other objects around it, and this applies as well to objects on the surface of the Earth. This is evident when an object is released from a certain height and undergoes free fall, meaning it is being pulled by the Earth. If there is no air resistance, the Earth's pull causes the object to experience an acceleration equal to the Earth's gravitational acceleration (g). Of course, the value of g depends on the distance from that location to the Earth's center and also on the material present beneath the Earth's surface at the local site.

Since the local value of g depends on the material beneath the Earth's surface, the presence of g can be utilized as an initial indicator of the existence of distinctive substances within the Earth, such as basins containing fossil water, hydrocarbons (petroleum), or metallic materials. However, there are various methods available for determining g . Each technique and method has its own advantages and disadvantages.

2.1 Objectives

This practical exercise has the following two objectives:

1. To determine the value of g using three methods: the mathematical pendulum, spring oscillation, and water oscillation in a U-tube;

2. To compare the three measurement methods in terms of accuracy and precision.

Note: If time is considered insufficient, two methods may be selected instead, namely: the mathematical pendulum and spring oscillation, or the mathematical pendulum and water oscillation in the U-tube.

2.2 Tools and Materials

Since this practical topic involves three methods for measuring g , it therefore requires three set-ups (equipment systems). These three set-ups are: the mathematical pendulum system, the spring oscillation system (stand, spring, and load), and the U-tube system. Each of these equipment set-ups requires supporting tools in order to become functional; these supporting tools are: a stopwatch (to measure the period) and a measuring tape (to measure the length of the string, the spring displacement, and the length of the fluid in the U-tube).

2.3 Theoretical Foundation

Since this experiment involves three methods, the theoretical foundation for each method will also be presented. Nevertheless, all three are based on the same principle, namely simple harmonic motion (SHM). SHM is an oscillatory motion (i.e., back-and-forth movement around its equilibrium point) that involves only one force, the restoring force (F_r). If the object undergoing SHM has a mass m and at t has a displacement x , the following condition is satisfied:

$$m \frac{d^2 x}{dt^2} = F_r \quad (2.1)$$

If the simple harmonic motion (SHM) has an angular frequency ω and displacement θ , then the SHM equation is:

$$\frac{d^2\theta}{dt^2} + \omega^2\theta = 0 \quad (2.2)$$

If the amplitude of the displacement is θ_0 and the phase constant is ϕ , then the solution to equation (2.2) is:

$$\theta(t) = \theta_0 \cos(\omega t + \phi) \quad (2.3)$$

If the experimental method does not meet the requirements of SHM, then an experimental method is chosen such that the correction factor is minimum.

2.3.1 Simple Pendulum

The existence of a simple pendulum is shown in Figure 2.1(a). It depicts a pendulum using a mass m attached to a string of length l and a pendulum displacement θ . In this case, the local gravitational acceleration is g , resulting in a pendulum period T . The pendulum displacement is $x = l \sin \theta$, while the restoring force is $F_r = -mg \sin \theta$.

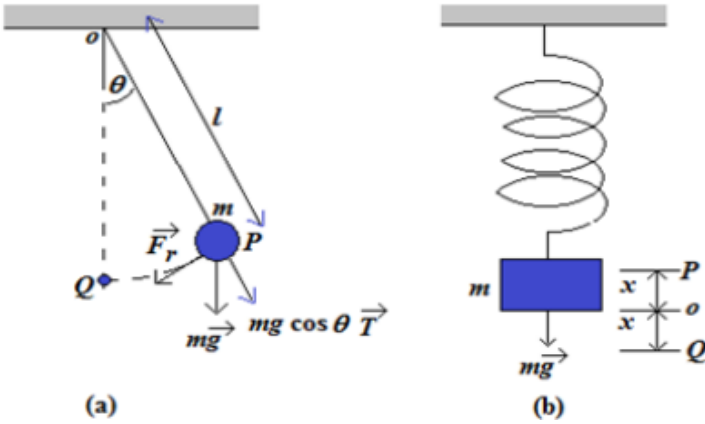


Figure 2.1: Chart: (a) simple pendulum (b) vibration of a spring

If the pendulum is taken under special conditions, namely with a small angle θ , then it satisfies $x = l\theta$ and $F_r = -mg\theta$. In addition, if the swinging object does not rotate and experiences no damping force (air resistance is negligible), according to equation 2.1: $ml(d^2\theta/dt^2) + mg\theta = 0$, or

$$\frac{d^2\theta}{dt^2} + \left(\frac{g}{l}\right)\theta \quad (2.4)$$

That equation is equivalent to equation 2.2, and thus the solution is the same as in equation 2., it's just that $\omega^2 = \frac{g}{l}$.

Since $\omega = \frac{2\pi}{T}$, then obtain:

$$T = 2\pi \sqrt{\frac{l}{g}} \quad (2.5)$$

Equation 2.5, is the fundamental basic for determining g based on the measured period T for each string length l .

2.3.2 Spring Vibration

In this spring vibration, the restoring force is a spring force. Thus, for a spring constant k and simple harmonic motion (SHM) with displacement x of a load (mass m), thus we have $F_r = -kx$, from equation 2.1 gives $m \left(\frac{d^2x}{dt^2} \right) + kx = 0$ or

$$\frac{d^2x}{dt^2} + \left(\frac{k}{m} \right) x = 0 \quad (2.6)$$

This equation is equivalent to equation 2.2, namely $\left(\frac{d^2\theta}{dt^2} \right) + \omega^2\theta = 0$ by correspondence $\omega^2 = k/m$, the displacement equation (with displacement amplitude x_0) is equivalent to equation 2.3, resulting in:

$$x = x_0 \cos(\omega t + \varphi) \quad (2.7)$$

In equation 2.6 and 2.7 it implies that for the period T , it satisfies:

$$\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}} \quad (2.8)$$

Since the spring is still at rest and given a load with a weight of mg , it extends by a length X . That means that the weight of the load is equal to the restoring force of the spring or $mg = kX$, so if this relation is combined with equation 2.8, the equation is obtained:

$$T = 2\pi \sqrt{\frac{X}{g}} \quad (2.9)$$

2.3.3 Oscillation of liquid in a U-tube

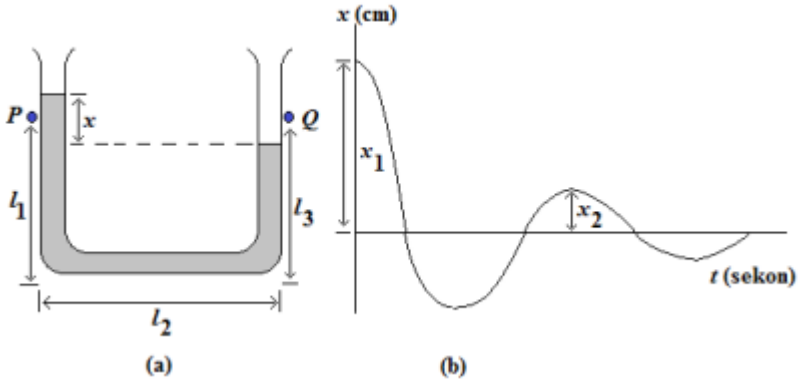


Figure 2.2: (a) Chart oscillation of liquid in a U-tube, (b) the oscillation of the water “actually” included damped harmonized vibrations

U-tube (Figure 2.2(a)) has a cross-sectional area A and contains a liquid of mass m , of density ρ , inside U-tube of length L . Thus, the relationship $m = \rho AL$ is satisfied. In the case, the liquid oscillates at displacement of $2x$ on both sides, at the local earth's gravitational acceleration g . Therefore, resulting in the restoring force:

$$F_r = -2x\rho Ag \quad (2.10)$$

If the equation is associated with equation 2.1, then the equation takes the form $\rho AL(d^2x/dt^2) + 2gx = 0$, or

$$\frac{d^2x}{dt^2} + \left(\frac{2g}{L}\right)x = 0 \quad (2.11)$$

Equation 2.11 is equivalent to equation 2.2, which means $\omega^2 = 2g/L$. The displacement is equivalent to equation 2.3, at the amplitude displacement of x_0 , and gives:

$$x = x_0 \cos(\omega t + \varphi) \quad (2.12)$$

Using $\omega = 2\pi/T = \sqrt{2g/L}$, or

$$T = 2\pi \sqrt{\frac{L}{2g}} \quad (2.13)$$

Actually, the oscillation in the U-tube is a Damped Harmonic Motion (DHM) because it is clearly observed that the amplitude of the oscillation decreases overtime, the decay pattern is visibly apparent in Figure 2.2(b). To enable the application of equation 2.11 (representing simple harmonic motion, SHM), so that equation 2.13 can be used to determine g , two approaches may be employed: **method 1** (practical but less precise) and **method 2** (less practical but more precise).

Method 1

It is chosen when the damping is minimized, i.e., when the length of liquid ($L = l_1 + l_2 + l_3$) minimized, while maintaining a sufficient number of oscillations. Equation 2.13 is then applied.

Method 2

The presence of damping effect is taken into account to correct the period value stated in equation 2.13. If the period of the

damped harmonic motion (DHM) is T_d and the damping constant of the oscillation is γ , then the n -th maximum displacement involving constant A at time $t = nT_d$ is $x_n = Ae^{-\gamma nT_d}$. In addition, the displacement at the $(n + 1)$ -th is $x_{n+1} = Ae^{-\gamma(n+1)T_d}$. If $x_n = ax_{n+1}$ ($a > 1$), then the ratio is obtained: $\frac{x_n}{x_{n+1}} = a = e^{\gamma T_d}$, or $\ln a = \gamma T_d$, which means:

$$\gamma = \frac{1}{T_d} \ln a \quad (2.14)$$

At the same time, the relationship between the damped angular frequency ($\omega_d = 2\pi/T_d$) and the undamped angular frequency $\omega_d = 2\pi/T$ is given by:

$$\omega_d^2 = \omega_0^2 - \gamma^2 \quad (2.15)$$

The combination of equation 2.15 with equation 2.14 gives the equation:

$$T = \frac{T_d}{\sqrt{1 + \left(\ln \frac{a}{2\pi}\right)^2}} \quad (2.16)$$

The period in equation 2.13 is treated as T_d and in order to be treated as a simple harmonic motion (SHM), it must be corrected using equation 2.16.

2.4 Experimental Methods

Since this experiment involves three experimental methods, namely the simple pendulum, spring vibration, and oscillation of liquid in a U-tube), the following describes these three methods.

2.4.1 Simple Pendulum

Experimental strategy

This method uses the basis of equation 2.5, which states that $T = 2\pi(l/g)$, the conditions required are that the mass (iron ball) does not rotate, the angular displacement is small (maximum of 10°), such that when the string length $l = 100\text{ cm}$, the maximum displacement is 17 cm . In order for the value of g to be measured accurately, both l and the period T must be measured precisely. To eliminate the effect of systematic error in l , a graph of T^2 vs l is plotted, rather than T vs $\sqrt{l}$. To make the measurement of T more practical, larger values of l are chosen, as they correspond to larger values of T , since $T \propto \sqrt{l}$. To improve the accuracy of the period, the time is recorded over multiple oscillations (e.g., every 10 oscillations). Furthermore, when the start and stop on the stopwatch at the most sensitive position, so that the initial and final moments of period measurement occurs at the equilibrium position (zero displacement). The apparatus used is shown in Figure 2.3.

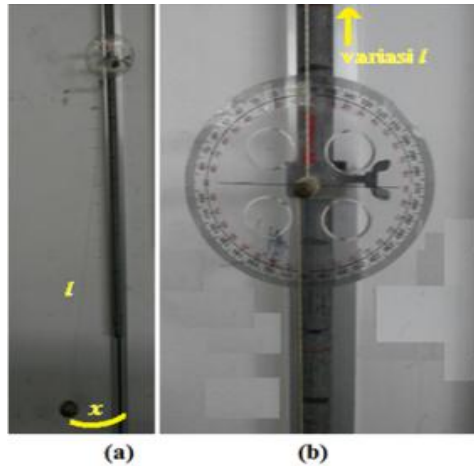


Figure 2.3: Photograph of a simple pendulum: (a) minimum string length of 100 cm and maximum x is 17 cm , (b) adjustment mechanism for string length and swing angle.

Procedure

1. Choose a string length of 100 cm , displace the pendulum to a maximum of 17 cm , and release the pendulum load (iron ball) so that it swings without rotation. Measure the oscillation period by timing every 10 oscillations, and record the period measurement result using a stopwatch. Repeat the period measurement once more. Note: the period measurement should start and stop when the pendulum load (iron ball) passes through its equilibrium position.
2. Perform step 1 for string length of: 100 cm , 120 cm , 130 cm , ..., up to 200 cm . If it is difficult to vary the string length for all these values, use a least 6 variations.

3. Present the experimental result in Table 2.1
4. Based on the equation $T^2 = (4\pi^2/g)l$, plot T^2 vs l , and using the best-fit slope along with its uncertainty, calculate the value of g , resulting in $g = \bar{g} \pm \Delta g$.

Table 2.1 Experimental result of a simple pendulum

Note: $\bar{T} = \frac{10T_1 + 10T_2}{20}$; $\Delta\bar{T} = \frac{|10T_1 - 10T_2|}{20}$; $\bar{T}^2 = (\bar{T})(\bar{T})$; dan $\Delta\bar{T}^2 = 2\bar{T}\Delta\bar{T}$

No	l (cm)	$10T$ (sekon)		$\bar{T} \pm \Delta\bar{T}$ (sekon)	$\bar{T}^2 \pm \Delta\bar{T}^2$ (x10 ³ sekon)
1	100				
2	110				
3	120				
10	200				

2.4.2 Spring Vibration

Experimental strategy

This method is based on equation 2.9, namely $T = 2\pi\sqrt{X/g}$. Naturally, the value of g can be obtained accurately if the system oscillates close to SHM, and both X and T are measured precisely. The spring system oscillates close to SHM when the air friction force is minimum. It happens when the cross-sectional area of the load mass is small and the motion is slow. To make the motion m slow, so that a large value of m is selected, but ensure that the spring elongation X does not exceed the length of the spring when unloaded. The reason is that when the spring elongation exceeds the spring length, Hooke's law ($F_r = -kX$) no longer valid. Furthermore, to eliminate the influence of systematic error on X in determining g , the equation is modified to $T^2 = (4\pi^2/g)X$ and a graph of

T^2 vs X is plotted. Naturally, to make the measurement of T easier, a large value of m is chosen so that X is also large, and T can be measured accurately by measuring every 10 periods. It should be noted that the measurement of T (using a stopwatch) starts and ends when m is at its equilibrium position (o in Figure 2.1b). The equipment used for this spring vibration experiment is shown in Figure 2.4.

Procedure

1. Take a spring and measure its length. Then, hang or clamp the upper end of the spring on a stative, while placing a mass hanger at the lower end. Add a load mass such that the elongation of the spring is approximately equal to the spring's length before stretching, and this load mass is referred to as the maximum load mass (m). If the maximum mass is 1600 gram, to obtain at least 6 variations, then vary the load masses as: 600; 800; 1.000; ...; dan 1600 gram.

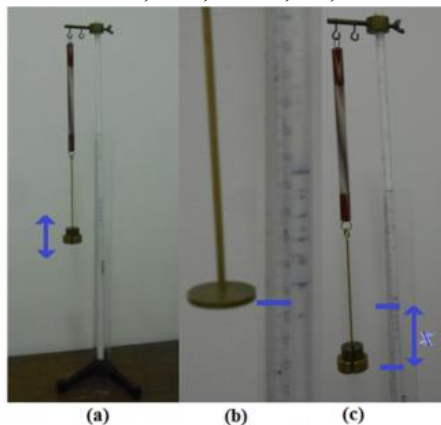


Figure 2.4: Photograph of the local gravitational acceleration measuring instrument using the spring

vibration method: (a) complete, (b) without load, and (c) with load.

- Place a 600-gram load on the hanger, and measure the elongation length (X). Then, oscillate the load (by pulling down the mass m and releasing it) and measure the periode (T) every 10 oscillations. Take that measurement twice, and remember the stopwatch should start at 0 and stop (after 10 oscillations) also at 0.
- Perform step b for load masses of 800, 1.000, ..., and 1.600 grams. Then, record the experimental results in Table 2.2.
- Based on the equation $T^2 = (4\pi/g)X$, plot T^2 vs X , and from the resulting slope and its uncertainty, calculate g to obtain the result in the form $g = (\bar{g} \pm \Delta g)$.

Table 2.2 Experimental result of a spring vibration

Note: $\bar{T} = \frac{10T_1 + 10T_2}{20}$; $\Delta\bar{T} = \frac{|10T_1 - 10T_2|}{20}$; $\bar{T}^2 = (\bar{T})(\bar{T})$; dan $\Delta\bar{T}^2 = 2\bar{T}\Delta\bar{T}$

No	X (cm)	$10T$ (sekon)	$\bar{T} \pm \Delta\bar{T}$ (sekon)	$\bar{T}^2 \pm \Delta\bar{T}^2$ (x10 ³ sekon)
1	600			
2	800			
3	1.000			
4	1.200			
5	1.400			
6	1.600			

2.4.3 Oscillation of liquid in a U-tube

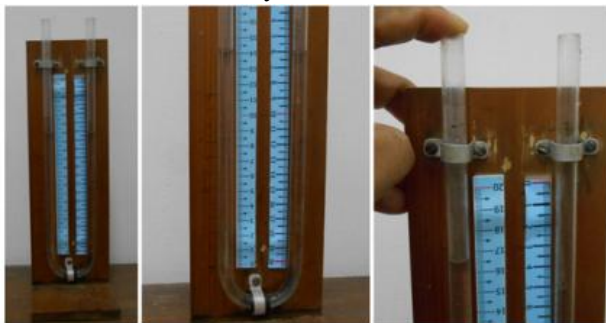
Experimental strategy

The experiment uses clear water as the liquid medium, and the experimental method is based on equation 2.13, namely $T =$

$2\pi\sqrt{L/2g}$, along with its correction expressed in equation 2.16, when the period of pure simple harmonic motion (SHM) is T_0 and the damped is T_d , so that $T_0 = T_d/\sqrt{1 + (\ln a/2\pi)^2}$. In addition, determine the ratio of the first displacement amplitude (x_1) with the second displacement amplitude (x_2) denoted as $x_1/x_2 = a$. Additionally, each measured the period value must be multiplied by $[1 + (\ln a/2\pi)^2]^{-\frac{1}{2}}$. To measure the period easily, select a mass m that result in a large period T , and to ensure sufficiently small damping, a short liquid length (L) is also selected. The U-tube apparatus is shown in Figure 2.5.

Procedure

1. Choose a liquid length ($L = l_1 + l_2 + l_3$) of 10 cm and oscillate it, and measuring the period every 4 oscillations. Repeat the period measurement once more.
2. Perform step 1 with a variation of 2 cm each time (by adding $l_1 = 1$ cm to the left leg of the U-tube and adding $l_2 = 1$ cm to the right leg of the U-tube), so that the subsequent values of L are respectively: 12, 14, 16, 18, and 20 cm. Record your data in table.



- Figure 2.5:** (a) Photograph of the oscillation of liquid in a U-tube, (b) before use, and (c) during oscillation
3. Based on the obtainable slope and the uncertainty slope of the T^2 vs L graph, determine the value of g so that the result is obtained as $g = (\bar{g} \pm \Delta g)$.